

Generation Interconnection System Impact Study Report

For Queue Project Q12-0143

Mickens 138 kV

August 18, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0143. The Interconnection Customer proposes a 1014.5 MW natural gas combined cycle facility interconnecting at the Mickens 138 kV substation in the Virginia area. The request was received on July 13, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 93 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$38,776,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Mickens / Yadkin 138 KV
Contingency: Swan / Mickens 138 KV
Loading: 311.3% (Rating: 312 MVA)
DFAX: 57.4%
Pre-project loading: 124.4%
2. Facility: Churchland / Benton City 138 KV
Contingency: Yadkin / Mickens 138 KV
Loading: 221.3% (Rating: 174 MVA)
DFAX: 26.9%
Pre-project loading: 64.3%
3. Facility: Swan / Mickens 138 KV L
Contingency: Yadkin / Mickens 138 KV
Loading: 203.9% (Rating: 247 MVA)
DFAX: 26.9%
Pre-project loading: 93.5%
4. Facility: Van Horne / Pt Roberts 138 KV
Contingency: Mickens / Yadkin 138 KV
Loading: 202.0% (Rating: 227 MVA)
DFAX: 26.9%
Pre-project loading: 81.9%
5. Facility: Van Horne / Swan 138 KV
Contingency: Mickens / Yadkin 138 KV
Loading: 201.6% (Rating: 241 MVA)
DFAX: 26.9%
Pre-project loading: 88.1%
6. Facility: Valley Farms / Pt Roberts Substation 138 KV
Contingency: Yadkin / Mickens 138 KV
Loading: 201.3% (Rating: 218 MVA)

DFAX: 26.9%

Pre-project loading: 75.9%

7. Facility: Valley Farms / Benton City 138 KV

Contingency: Yadkin / Mickens 138 KV

Loading: 200.0% (Rating: 206 MVA)

DFAX: 26.9%

Pre-project loading: 67.6%

8. Facility: Harborgen / Yadkin 138 KV

Contingency: Yadkin / Wheelabrator Portsmouth 138 KV

Loading: 198.8% (Rating: 222 MVA)

DFAX: 25.5%

Pre-project loading: 82.2%

9. Facility: Wheelabrator Portsmouth / Yadkin 138 KV

Contingency: Harborgen / Yadkin 138 KV Line

Loading: 164.8% (Rating: 334 MVA)

DFAX: 30.4%

Pre-project loading: 72.3%

10. Facility: Churchland / Trans Bay Cable Facility 138 KV L

Contingency: Yakin / Mickens 138 KV

Loading: 150.3% (Rating: 221 MVA)

DFAX: 26.9%

Pre-project loading: 26.6%

11. Facility: Trans Ba Cable Facility / Gosport 138 KV

Contingency: Mickens / Yadkin 138 KV

Loading: 128.7% (Rating: 210 MVA)

DFAX: 15.1%

Pre-project loading: 56.1%

12. Facility: Wheelabrator Portsmouth / Gosport 138 KV Line

Contingency: Micens / Swan 138 KV

Loading: 127.2% (Rating: 254 MVA)

DFAX: 14.2%

Pre-project loading: 70.7%

13. Facility: Unknown / N Tower Tap 138 KV

Contingency: N Tower Tap / Haggin 345 KV L

Loading: 105.0% (Rating: 277 MVA)

DFAX: 8.0%

Pre-project loading: 75.6%

14. Facility: Walthall / Unknown 138 KV

Contingency: N Tower Tap / Haggin 345 KV

Loading: 101.7% (Rating: 272 MVA)

DFAX: 8.0%

Pre-project loading: 71.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yadkin / Mickens 138 KV	Rebuild line with higher-capacity conductor	\$2,254,000	93% (\$2,099,000)
Churchland / Benton City 138 KV	Rebuild line with higher-capacity conductor	\$2,426,000	100% (\$2,426,000)
Mickens / Swan 138 KV	Rebuild line with higher-capacity conductor	\$5,011,000	100% (\$5,011,000)
Van Horne / Point Roberts 138 KV	Rebuild line with higher-capacity conductor	\$2,049,000	100% (\$2,049,000)
Swan / Van Horne 138 KV L	Rebuild line with higher-capacity conductor	\$1,742,000	100% (\$1,742,000)
Vly Farms / Point Roberts Substation 138 KV	Rebuild line with higher-capacity conductor	\$3,900,000	100% (\$3,900,000)
Valley Farms / Benton City 138 KV L	Rebuild line with higher-capacity conductor	\$4,058,000	100% (\$4,058,000)
Yadkin / Harborgen 138 KV line	Rebuild line with higher-capacity conductor	\$3,267,000	81% (\$2,641,000)
Wheelabrator Portsmouth / Yadkn 138 KV	Rebuild line with higher-capacity conductor	\$7,276,000	80% (\$5,857,000)
Churchland / Trans Bay Cable Facility 138 KV L	Rebuild line with higher-capacity conductor	\$750,000	100% (\$750,000)
Trans Ba Cable Facility / Gosport 138 KV	Rebuild line with higher-capacity conductor	\$2,138,000	78% (\$1,663,000)
Wheelabrator Portsmouth / Gosprt 138 KV L	Rebuild line with higher-capacity conductor	\$1,546,000	81% (\$1,247,000)
Unknown / N Tower Tap 138 KV	Reconductor line	\$1,508,000	71% (\$1,068,000)
Walthall Substation / Unknown 138 KV	Reconductor line	\$6,019,000	71% (\$4,265,000)
	Total allocated to project		\$38,776,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.